Practical 3: ML in Cybersecurity – Comparative Case Study

Domain: Network Intrusion Detection

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Aim

Compare three machine-learning techniques used to identify malicious network traffic: Decision Tree, Random Forest, and Support Vector Machine (SVM).

Problem statement

A Network Intrusion Detection System (NIDS) examines connection-level features and predicts whether traffic is normal or associated with an attack. A model can reduce manual review, but false negatives may allow attacks through and false positives can overwhelm analysts.

Dataset

NSL-KDD is a widely used benchmark derived from KDD Cup 1999. Each connection contains 41 traffic/connection features and a label. It reduces some redundant-record problems of the original KDD'99 dataset, but it is old and does not represent all current network traffic. Dataset page: https://www.unb.ca/cic/datasets/nsl.html

Techniques compared

1. Decision Tree

A tree recursively splits the feature space using decision rules. It is interpretable and can model non-linear relationships. However, a deep tree may overfit and can be unstable when the data change.

2. Random Forest

A Random Forest trains many decision trees on bootstrapped samples and randomized feature subsets, then combines their predictions. It generally reduces the variance of a single tree and captures feature interactions. Its disadvantages include higher compute cost and less direct interpretability.

3. Support Vector Machine (SVM)

An SVM seeks a decision boundary with a large margin between classes. Kernel SVMs can model non-linear boundaries; linear SVMs can be efficient on high-dimensional encoded features. Scaling is important, and training can be expensive on large datasets. Probability estimates are not automatically available in all SVM configurations.

Published evidence (reported values are study-specific)

A 2026 NSL-KDD study reported accuracy of **81.58% for XGBoost, 81.46% for Random Forest, and 81.43% for Decision Tree** after its particular preprocessing and ADASYN balancing pipeline. These figures illustrate that tree ensembles and individual trees can be competitive, but they are not universal benchmarks and are not directly comparable to results from a different split or preprocessing pipeline. Source: https://www.sciencedirect.com/science/article/pii/S1877050926022052

A 2016 study evaluating supervised methods on NSL-KDD reported that Random Forest outperformed the other methods it tested, including SVM, for its experimental setup. Source: https://www.sciencedirect.com/science/article/pii/S187705091631081X

A 2024 IEEE conference study compared traditional ML and deep-learning methods on NSL-KDD and UNSW-NB15 using accuracy, precision, recall, and F1-score, emphasizing that performance depends on both model and dataset. Source: https://ieeexplore.ieee.org/abstract/document/10625037

Comparison table

Technique	Main strength	Main limitation	Suitable use
Decision Tree	Explainable rules; quick inference	Can overfit and be unstable	Baseline model and analyst explanation
Random Forest	Strong general-purpose performance; lower variance than one tree	More memory and less transparent	Robust baseline for tabular traffic
SVM	Effective high-dimensional decision boundaries	Scaling and tuning matter; large datasets can be costly	Smaller/medium datasets or linear high-dimensional features

Evaluation metrics

- **Accuracy:** proportion of all predictions that are correct.
- **Precision:** proportion of predicted attacks that are actually attacks.
- **Recall:** proportion of actual attacks detected.
- **F1-score:** harmonic mean of precision and recall.
- **False-positive rate:** benign traffic incorrectly flagged as malicious.

For intrusion detection, recall and false-positive rate are especially important. A high accuracy score can be misleading when classes are imbalanced.

Conclusion

Decision Tree, Random Forest, and SVM offer different trade-offs. Decision Trees are easier to explain, Random Forests often provide a strong balance of performance and robustness, and SVMs can work well with scaled high-dimensional data. A fair comparison must use the same dataset split, preprocessing, and metrics. Benchmark results should not be assumed to transfer directly to live networks.

References

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